

Eigenvalues & Eigenfunctions

*Characteristic Equations, Eigenvalues & Eigenvectors,
and the Fundamental Matrix for Linear Systems*

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Eigenvalue Problem • Characteristic Equation • Eigenvalues • Eigenvectors • Complex &
Repeated Cases • Eigenfunction Solutions • Fundamental Matrix • $\Phi(t)$ with $\Phi(0) = I$

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1 Part I | Condensed Cheat Sheet

1.1 The Eigenvalue Problem

For a square matrix A , most vectors are rotated *and* stretched when multiplied by A . A few special directions are only stretched — those are the eigenvectors, and the stretch factors are the eigenvalues. This single idea unlocks the solution of every constant-coefficient linear system $\mathbf{x}' = A\mathbf{x}$, because along an eigenvector the matrix behaves like a scalar.

Definition. A scalar λ and *nonzero* vector $\boldsymbol{\xi}$ form an *eigenpair* of the $n \times n$ matrix A when

$$A\boldsymbol{\xi} = \lambda\boldsymbol{\xi} \quad \Longleftrightarrow \quad (A - \lambda I)\boldsymbol{\xi} = \mathbf{0}.$$

A nontrivial $\boldsymbol{\xi}$ exists exactly when $A - \lambda I$ is singular.

1.2 The Characteristic Equation

Singularity of $A - \lambda I$ is detected by its determinant — setting it to zero produces a degree- n polynomial whose roots are the eigenvalues.

Characteristic equation:

$$\det(A - \lambda I) = 0.$$

2×2 shortcut. For any 2×2 matrix,

$$\lambda^2 - (\operatorname{tr} A)\lambda + \det A = 0,$$

and the roots satisfy $\lambda_1 + \lambda_2 = \operatorname{tr} A$ and $\lambda_1\lambda_2 = \det A$ — a built-in check.

1.3 Finding Eigenvectors

For each eigenvalue λ , the matrix $A - \lambda I$ is singular, so $(A - \lambda I)\boldsymbol{\xi} = \mathbf{0}$ has a free variable. The eigenvector is therefore determined only *up to a scalar multiple* — choose a convenient scaling.

Worked Example — real distinct eigenvalues. Find the eigenpairs of $A = \begin{pmatrix} 5 & -1 \\ 3 & 1 \end{pmatrix}$.

Step 1. Characteristic equation via the shortcut ($\operatorname{tr} A = 6$, $\det A = 8$):

$$\lambda^2 - 6\lambda + 8 = 0 \implies (\lambda - 2)(\lambda - 4) = 0 \implies \lambda = 2, 4.$$

Step 2. Eigenvector for $\lambda = 2$ — solve $(A - 2I)\boldsymbol{\xi} = \mathbf{0}$:

$$\begin{pmatrix} 3 & -1 \\ 3 & -1 \end{pmatrix} \boldsymbol{\xi} = \mathbf{0} \implies 3\xi_1 - \xi_2 = 0 \implies \boldsymbol{\xi}^{(1)} = \begin{pmatrix} 1 \\ 3 \end{pmatrix}.$$

Step 3. Eigenvector for $\lambda = 4$ — solve $(A - 4I)\boldsymbol{\xi} = \mathbf{0}$:

$$\begin{pmatrix} 1 & -1 \\ 3 & -3 \end{pmatrix} \boldsymbol{\xi} = \mathbf{0} \implies \xi_1 - \xi_2 = 0 \implies \boldsymbol{\xi}^{(2)} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

1.4 From Eigenpairs to Solutions (the “Eigenfunctions”)

Each eigenpair generates an exponential solution of $\mathbf{x}' = A\mathbf{x}$ — these are the eigenfunctions of the linear system. Three cases cover every 2×2 matrix.

Basic solution. Each eigenpair $(\lambda, \boldsymbol{\xi})$ gives $\mathbf{x}(t) = \boldsymbol{\xi} e^{\lambda t}$.

Real distinct $\lambda_1, \dots, \lambda_n$:

$$\mathbf{x}(t) = c_1 \boldsymbol{\xi}^{(1)} e^{\lambda_1 t} + \dots + c_n \boldsymbol{\xi}^{(n)} e^{\lambda_n t}.$$

Complex pair $\lambda = \alpha \pm i\beta$, $\boldsymbol{\xi} = \mathbf{a} + i\mathbf{b}$: take real and imaginary parts of $\boldsymbol{\xi} e^{\lambda t}$:

$$\mathbf{u}(t) = e^{\alpha t}(\mathbf{a} \cos \beta t - \mathbf{b} \sin \beta t), \quad \mathbf{v}(t) = e^{\alpha t}(\mathbf{a} \sin \beta t + \mathbf{b} \cos \beta t).$$

Repeated λ (defective — only one eigenvector $\boldsymbol{\xi}$): find a generalized eigenvector $\boldsymbol{\eta}$ from $(A - \lambda I)\boldsymbol{\eta} = \boldsymbol{\xi}$, then

$$\mathbf{x}^{(1)} = \boldsymbol{\xi} e^{\lambda t}, \quad \mathbf{x}^{(2)} = (\boldsymbol{\xi} t + \boldsymbol{\eta}) e^{\lambda t}.$$

Worked Example — complex eigenvalues. Solve the eigenproblem for $A = \begin{pmatrix} -1 & -4 \\ 1 & -1 \end{pmatrix}$ and build two real solutions.

Step 1. Characteristic equation ($\text{tr } A = -2$, $\det A = 5$):

$$\lambda^2 + 2\lambda + 5 = 0 \implies \lambda = -1 \pm 2i, \quad \alpha = -1, \beta = 2.$$

Step 2. Eigenvector for $\lambda = -1 + 2i$ — from $(A - \lambda I)\boldsymbol{\xi} = \mathbf{0}$, the row $\xi_1 - 2i\xi_2 = 0$ gives

$$\boldsymbol{\xi} = \begin{pmatrix} 2i \\ 1 \end{pmatrix} = \underbrace{\begin{pmatrix} 0 \\ 1 \end{pmatrix}}_{\mathbf{a}} + i \underbrace{\begin{pmatrix} 2 \\ 0 \end{pmatrix}}_{\mathbf{b}}.$$

Step 3. Real solutions via the real/imaginary parts:

$$\mathbf{u}(t) = e^{-t} \begin{pmatrix} -2 \sin 2t \\ \cos 2t \end{pmatrix}, \quad \mathbf{v}(t) = e^{-t} \begin{pmatrix} 2 \cos 2t \\ \sin 2t \end{pmatrix}.$$

Because $\alpha = -1 < 0$, every trajectory spirals inward — a stable spiral.

Worked Example — repeated (defective) eigenvalue. Solve for $A = \begin{pmatrix} 3 & -1 \\ 1 & 1 \end{pmatrix}$.

Step 1. Characteristic equation ($\text{tr } A = 4$, $\det A = 4$):

$$\lambda^2 - 4\lambda + 4 = (\lambda - 2)^2 = 0 \implies \lambda = 2 \text{ (double)}.$$

Step 2. Only one eigenvector — $(A - 2I)\boldsymbol{\xi} = \mathbf{0}$ gives $\xi_1 - \xi_2 = 0$:

$$\boldsymbol{\xi} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Step 3. Generalized eigenvector from $(A - 2I)\boldsymbol{\eta} = \boldsymbol{\xi}$, i.e. $\eta_1 - \eta_2 = 1$:

$$\boldsymbol{\eta} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

Step 4. The two independent solutions:

$$\mathbf{x}^{(1)} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{2t}, \quad \mathbf{x}^{(2)} = \left[\begin{pmatrix} 1 \\ 1 \end{pmatrix} t + \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right] e^{2t}.$$

1.5 The Fundamental Matrix

Stacking n independent solutions side by side as columns produces the fundamental matrix — a single object that packages the entire solution space.

Fundamental matrix: $\Psi(t) = [\mathbf{x}^{(1)}(t) \mid \cdots \mid \mathbf{x}^{(n)}(t)]$, with independent columns. Key properties:

- $\Psi'(t) = A \Psi(t)$, and $\det \Psi(t) \neq 0$ for all t (Abel's theorem).
- General solution: $\mathbf{x}(t) = \Psi(t) \mathbf{c}$.
- IVP $\mathbf{x}(t_0) = \mathbf{x}^0$: $\mathbf{x}(t) = \Psi(t) \Psi(t_0)^{-1} \mathbf{x}^0$.

Special fundamental matrix $\Phi(t) = \Psi(t) \Psi(t_0)^{-1}$ satisfies $\Phi(t_0) = I$; then $\mathbf{x}(t) = \Phi(t) \mathbf{x}^0$, and $\Phi(t) = e^{A(t-t_0)}$ (the matrix exponential).

Worked Example — building $\Psi(t)$ and $\Phi(t)$. Use the eigenpairs of $A = \begin{pmatrix} 5 & -1 \\ 3 & 1 \end{pmatrix}$: $(2, (1, 3)^\top)$ and $(4, (1, 1)^\top)$.

Step 1. Columns are the two eigenfunction solutions:

$$\Psi(t) = \begin{pmatrix} e^{2t} & e^{4t} \\ 3e^{2t} & e^{4t} \end{pmatrix}, \quad \det \Psi = e^{6t}(1 - 3) = -2e^{6t} \neq 0.$$

Step 2. Evaluate at $t = 0$ and invert:

$$\Psi(0) = \begin{pmatrix} 1 & 1 \\ 3 & 1 \end{pmatrix}, \quad \Psi(0)^{-1} = \frac{1}{-2} \begin{pmatrix} 1 & -1 \\ -3 & 1 \end{pmatrix} = \begin{pmatrix} -\frac{1}{2} & \frac{1}{2} \\ \frac{3}{2} & -\frac{1}{2} \end{pmatrix}.$$

Step 3. Form $\Phi(t) = \Psi(t) \Psi(0)^{-1}$ (so that $\Phi(0) = I$):

$$\Phi(t) = \frac{1}{2} \begin{pmatrix} 3e^{4t} - e^{2t} & e^{2t} - e^{4t} \\ 3e^{4t} - 3e^{2t} & 3e^{2t} - e^{4t} \end{pmatrix}.$$

Note (Eigenvalues = Stability). The signs of the eigenvalues' real parts decide everything: all negative $\Rightarrow$ stable; any positive $\Rightarrow$ unstable; complex $\Rightarrow$ spiraling; pure imaginary $\Rightarrow$ a center. The same eigenvalues are the characteristic roots of the associated scalar ODE.

2 Part II | Review Guide

Ten problems, from a single characteristic equation up to a full fundamental-matrix IVP. Solutions follow in Part III.

1. Write the characteristic equation and find the eigenvalues of $A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$.
2. Find the eigenvalues of the symmetric matrix $A = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$.
3. Find the eigenvalues and eigenvectors of $A = \begin{pmatrix} 1 & 1 \\ 4 & 1 \end{pmatrix}$.
4. Find the eigenvalues and eigenvectors of $A = \begin{pmatrix} 2 & -1 \\ 3 & -2 \end{pmatrix}$.
5. Find the eigenvalues and a complex eigenvector of $A = \begin{pmatrix} 1 & -2 \\ 2 & 1 \end{pmatrix}$.
6. Using the eigenpair from Problem 5, write two real-valued solutions of $\mathbf{x}' = A\mathbf{x}$.
7. The matrix $A = \begin{pmatrix} 1 & -1 \\ 1 & 3 \end{pmatrix}$ has a repeated eigenvalue. Find it, its eigenvector, and a generalized eigenvector.
8. Find the characteristic equation and eigenvalues of $A = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 6 & -11 & 6 \end{pmatrix}$, then give an eigenvector for each.
9. For $A = \begin{pmatrix} 1 & 1 \\ 4 & 1 \end{pmatrix}$ (eigenpairs from Problem 3), write a fundamental matrix $\Psi(t)$ for $\mathbf{x}' = A\mathbf{x}$ and the special fundamental matrix $\Phi(t)$ with $\Phi(0) = I$.
10. Solve the IVP $\mathbf{x}' = \begin{pmatrix} 3 & -2 \\ 2 & -2 \end{pmatrix} \mathbf{x}$, $\mathbf{x}(0) = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, using eigenvalues and eigenvectors.

3 Part III | Answer Key & Worked Solutions

Solution 1

With $\text{tr } A = 7$, $\det A = 10$:

$$\lambda^2 - 7\lambda + 10 = 0 \implies (\lambda - 2)(\lambda - 5) = 0 \implies \lambda = 2, 5.$$

Solution 2

With $\text{tr } A = 6$, $\det A = 8$:

$$\lambda^2 - 6\lambda + 8 = 0 \implies (\lambda - 2)(\lambda - 4) = 0 \implies \lambda = 2, 4.$$

Solution 3

$\text{tr } A = 2$, $\det A = -3$:

$$\lambda^2 - 2\lambda - 3 = (\lambda - 3)(\lambda + 1) = 0 \implies \lambda = 3, -1.$$

For $\lambda = 3$: $(A - 3I)\boldsymbol{\xi} = \mathbf{0} \Rightarrow -2\xi_1 + \xi_2 = 0$:

$$\boldsymbol{\xi}^{(1)} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

For $\lambda = -1$: $(A + I)\boldsymbol{\xi} = \mathbf{0} \Rightarrow 2\xi_1 + \xi_2 = 0$:

$$\boldsymbol{\xi}^{(2)} = \begin{pmatrix} 1 \\ -2 \end{pmatrix}.$$

Solution 4

$\text{tr } A = 0$, $\det A = -1$:

$$\lambda^2 - 1 = 0 \implies \lambda = \pm 1.$$

For $\lambda = 1$: $(A - I)\boldsymbol{\xi} = \mathbf{0} \Rightarrow \xi_1 - \xi_2 = 0$:

$$\boldsymbol{\xi}^{(1)} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

For $\lambda = -1$: $(A + I)\boldsymbol{\xi} = \mathbf{0} \Rightarrow 3\xi_1 - \xi_2 = 0$:

$$\boldsymbol{\xi}^{(2)} = \begin{pmatrix} 1 \\ 3 \end{pmatrix}.$$

(Opposite-sign eigenvalues $\Rightarrow$ a saddle.)

Solution 5

$\text{tr } A = 2$, $\det A = 5$:

$$\lambda^2 - 2\lambda + 5 = 0 \implies \lambda = 1 \pm 2i.$$

For $\lambda = 1 + 2i$: $(A - \lambda I)\boldsymbol{\xi} = \mathbf{0}$ gives the row $-2i\xi_1 - 2\xi_2 = 0$, i.e. $\xi_2 = -i\xi_1$:

$$\boldsymbol{\xi} = \begin{pmatrix} 1 \\ -i \end{pmatrix}.$$

Solution 6

Write $\boldsymbol{\xi} = \mathbf{a} + i\mathbf{b}$ with $\mathbf{a} = (1, 0)^\top$, $\mathbf{b} = (0, -1)^\top$, and $\alpha = 1$, $\beta = 2$. Then

$$\mathbf{u}(t) = e^t(\mathbf{a} \cos 2t - \mathbf{b} \sin 2t) = e^t \begin{pmatrix} \cos 2t \\ \sin 2t \end{pmatrix},$$

$$\mathbf{v}(t) = e^t(\mathbf{a} \sin 2t + \mathbf{b} \cos 2t) = e^t \begin{pmatrix} \sin 2t \\ -\cos 2t \end{pmatrix}.$$

Since $\alpha = 1 > 0$, trajectories spiral outward — an unstable spiral.

Solution 7

$\text{tr } A = 4$, $\det A = 4$:

$$\lambda^2 - 4\lambda + 4 = (\lambda - 2)^2 = 0 \implies \lambda = 2 \text{ (double)}.$$

Eigenvector — $(A - 2I)\boldsymbol{\xi} = \mathbf{0} \Rightarrow -\xi_1 - \xi_2 = 0$:

$$\boldsymbol{\xi} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

Generalized eigenvector — $(A - 2I)\boldsymbol{\eta} = \boldsymbol{\xi} \Rightarrow -\eta_1 - \eta_2 = 1$, choose $\eta_1 = 0$:

$$\boldsymbol{\eta} = \begin{pmatrix} 0 \\ -1 \end{pmatrix}.$$

Solution 8

Expand $\det(A - \lambda I)$ along the first column:

$$\det \begin{pmatrix} -\lambda & 1 & 0 \\ 0 & -\lambda & 1 \\ 6 & -11 & 6 - \lambda \end{pmatrix} = -\lambda(\lambda^2 - 6\lambda + 11) + 6 = -\lambda^3 + 6\lambda^2 - 11\lambda + 6.$$

Setting this to zero:

$$\lambda^3 - 6\lambda^2 + 11\lambda - 6 = (\lambda - 1)(\lambda - 2)(\lambda - 3) = 0 \implies \lambda = 1, 2, 3.$$

This is a companion matrix, so each eigenvector has the form $(1, \lambda, \lambda^2)^\top$:

$$\boldsymbol{\xi}^{(1)} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \quad \boldsymbol{\xi}^{(2)} = \begin{pmatrix} 1 \\ 2 \\ 4 \end{pmatrix}, \quad \boldsymbol{\xi}^{(3)} = \begin{pmatrix} 1 \\ 3 \\ 9 \end{pmatrix}.$$

Solution 9

From Problem 3 the eigenfunction solutions are $(1, 2)^\top e^{3t}$ and $(1, -2)^\top e^{-t}$, so

$$\Psi(t) = \begin{pmatrix} e^{3t} & e^{-t} \\ 2e^{3t} & -2e^{-t} \end{pmatrix}.$$

Evaluate at $t = 0$ and invert:

$$\Psi(0) = \begin{pmatrix} 1 & 1 \\ 2 & -2 \end{pmatrix}, \quad \Psi(0)^{-1} = \frac{1}{-4} \begin{pmatrix} -2 & -1 \\ -2 & 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & \frac{1}{4} \\ \frac{1}{2} & -\frac{1}{4} \end{pmatrix}.$$

Then $\Phi(t) = \Psi(t)\Psi(0)^{-1}$ gives the matrix with $\Phi(0) = I$:

$$\Phi(t) = \begin{pmatrix} \frac{1}{2}(e^{3t} + e^{-t}) & \frac{1}{4}(e^{3t} - e^{-t}) \\ e^{3t} - e^{-t} & \frac{1}{2}(e^{3t} + e^{-t}) \end{pmatrix}.$$

Solution 10

$\text{tr } A = 1$, $\det A = -2$:

$$\lambda^2 - \lambda - 2 = (\lambda - 2)(\lambda + 1) = 0 \implies \lambda = 2, -1.$$

For $\lambda = 2$: $(A - 2I)\boldsymbol{\xi} = \mathbf{0} \Rightarrow \xi_1 - 2\xi_2 = 0 \Rightarrow \boldsymbol{\xi}^{(1)} = (2, 1)^\top$.

For $\lambda = -1$: $(A + I)\boldsymbol{\xi} = \mathbf{0} \Rightarrow 2\xi_1 - \xi_2 = 0 \Rightarrow \boldsymbol{\xi}^{(2)} = (1, 2)^\top$.

General solution and initial condition:

$$\mathbf{x}(t) = c_1 \begin{pmatrix} 2 \\ 1 \end{pmatrix} e^{2t} + c_2 \begin{pmatrix} 1 \\ 2 \end{pmatrix} e^{-t}, \quad \mathbf{x}(0) = \begin{pmatrix} 2c_1 + c_2 \\ c_1 + 2c_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

Solving $2c_1 + c_2 = 1$, $c_1 + 2c_2 = 0$ gives $c_1 = \frac{2}{3}$, $c_2 = -\frac{1}{3}$. Therefore

$$\mathbf{x}(t) = \frac{2}{3} \begin{pmatrix} 2 \\ 1 \end{pmatrix} e^{2t} - \frac{1}{3} \begin{pmatrix} 1 \\ 2 \end{pmatrix} e^{-t} = \begin{pmatrix} \frac{4}{3}e^{2t} - \frac{1}{3}e^{-t} \\ \frac{2}{3}e^{2t} - \frac{2}{3}e^{-t} \end{pmatrix}.$$